NOVEMBER 2026 · EXPERIMENTAL PROTOTYPE

Bench measurements

Two days of research: target responses, four complex channels over S/PDIF, and joint recording with a depth camera. These are observations from one prototype, not a detection-depth specification or a validated classifier.

Setup. The coil is horizontal, approximately 650 mm above a wooden floor over reinforced concrete. The operator reported an approximately 100 mm wooden layer; the detailed construction was not established. Targets moved between the floor and coil. Distances on 10 November were approximate operator estimates.

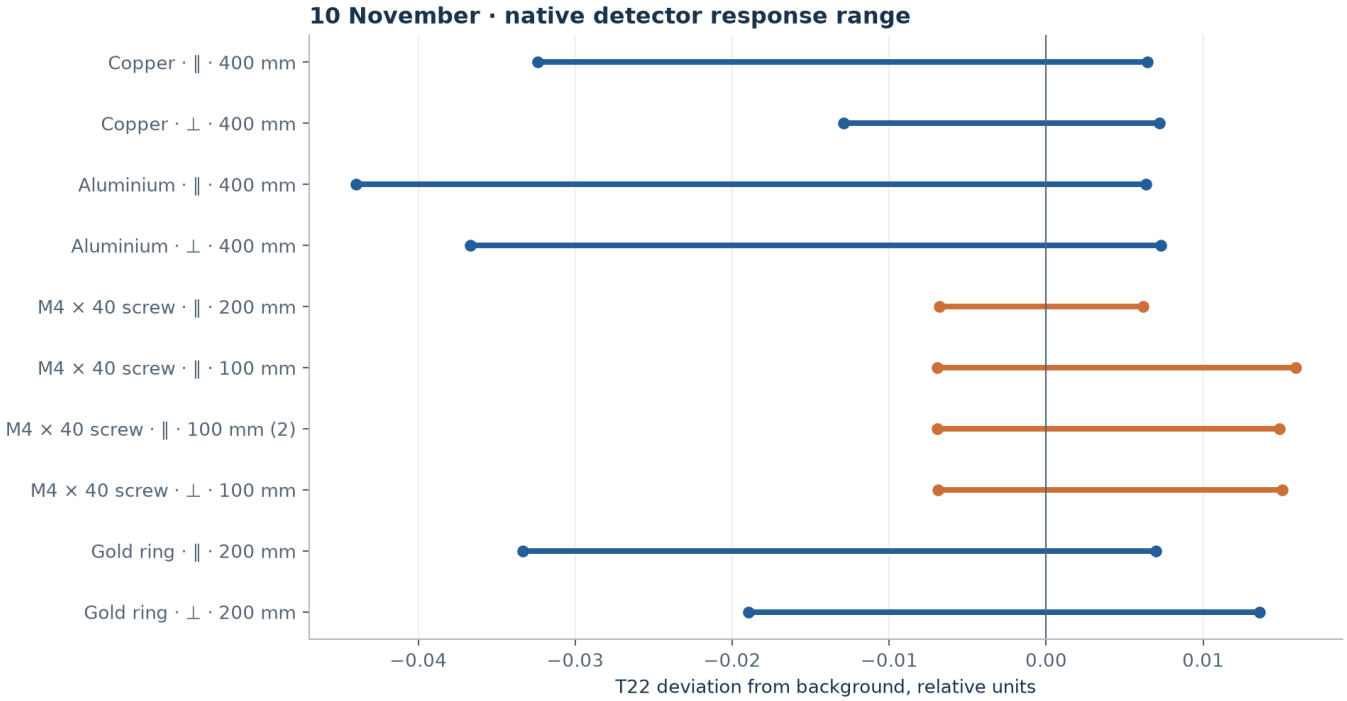
Equipment. FerroDSP: ADAU1452 (Analog Devices), two PCM1808 ADCs and PCM5102A DAC (Texas Instruments), TDA2050 TX amplifier (STMicroelectronics). Configuration: SigmaStudio 4.7 and USBi; the maker of the particular USBi adapter is unconfirmed. Audio capture: Creative Sound Blaster G3 optical input. Camera: Intel RealSense D415, USB, active stereo depth and color imaging. No separate video capture card was used.

Operating modes. DSP and audio capture run at 96 kHz; excitation is 15.2 kHz. Native phase project: TX -30 dB. Direct optical trials and PC Stream: TX -18 dB. These are digital generator levels, not power in watts. WAV uses a PCM24 container; this does not establish the effective resolution of the G3.

Main result. All four complex inputs are retained for algorithm development. RX0/RX1 interchannel phase did not yield a universal “sign = metal class” rule. Size, orientation and distance remain confounding factors.

10 NOVEMBER · USBI READBACK

Native phase response



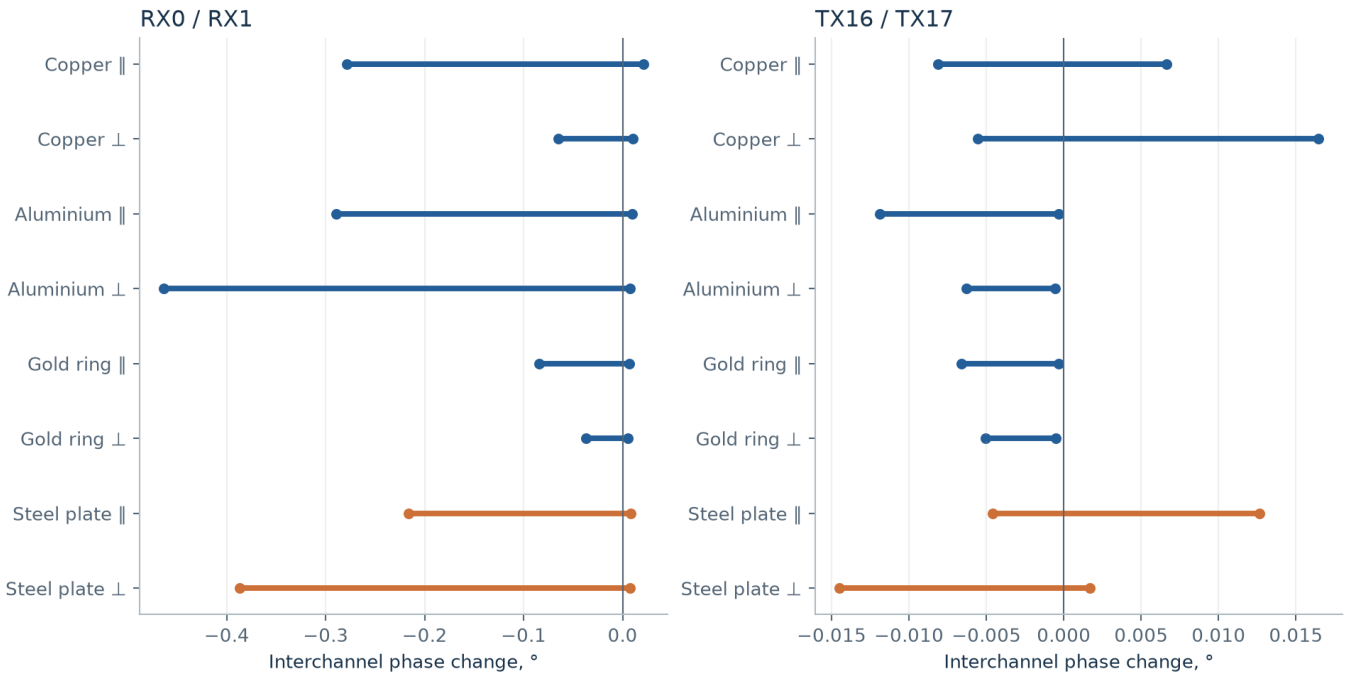
T22 is the signed, filtered balance-detector output in relative units; it is **not an angle in degrees**. Segments show the background-corrected minimum and maximum, not confidence intervals. ∥ / ⊥ refer to foil/ring planes; for the screw they refer to its axis relative to the coil plane.

Copper approximately 50 × 70 mm, aluminium 70 × 70 mm, an M4 × 40 mm screw and a gold wedding ring. Minima: copper ∥ at 400 mm -0.03239; aluminium ∥ at 400 mm -0.04396; ring ∥ at 200 mm -0.03332. Screw ∥ at 200 mm: -0.00678…+0.00619, approximately 3 background σ. At 100 mm the screw maximum reached +0.01589.

The graph contains 10 labeled recordings. One screw recording is excluded because its 200 mm distance and orientation were not confirmed. The screw_perpendicular_02 label was corrected from the operator’s clarification: axis parallel to the coil, 100 mm. These observations do not establish maximum detection depth.

10 NOVEMBER · SEPARATE RX AND TX PASSES

Direct optical recordings



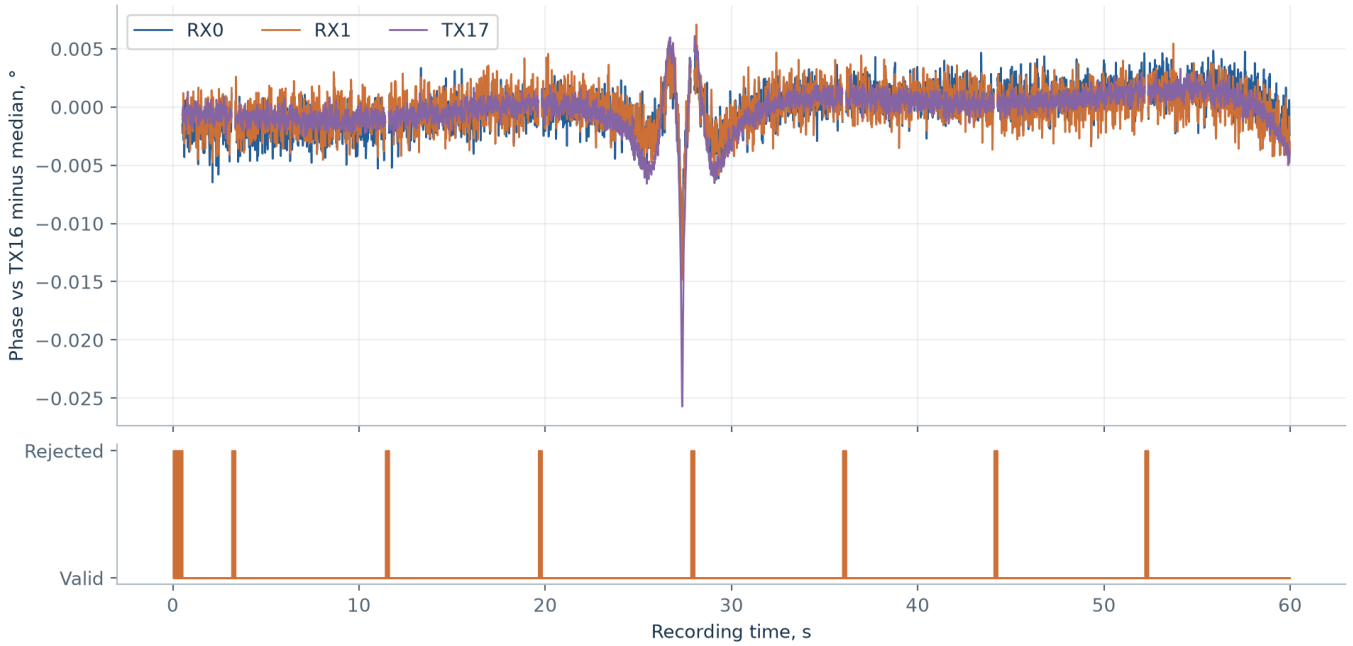
16 recordings: RX0/RX1 and TX16/TX17, four targets, two orientations. Foil and ring were approximately 200 mm away; the steel plate was approximately 400 mm away to reduce overload. Amplitudes across materials therefore do not represent identical geometry.

Processing: complex demodulation at 15.2 kHz, a 20 ms Hann window, 50 estimates/s. The ratio q = Z0/Z1 is normalized to the median background 12…2 s before the instruction. A five-sample median filter and bidirectional 3 Hz low-pass filter are used for inspection. This is offline processing. Bounds correspond to the instruction and reply, not exact physical pass times.

Finding: negative RX interchannel phase changes occurred for both nonferrous samples and steel. RX and TX were recorded sequentially; these trials cannot reconstruct simultaneous RX-to-TX phase.

FERRODSP-AM-IQ-V1

Four-channel transport



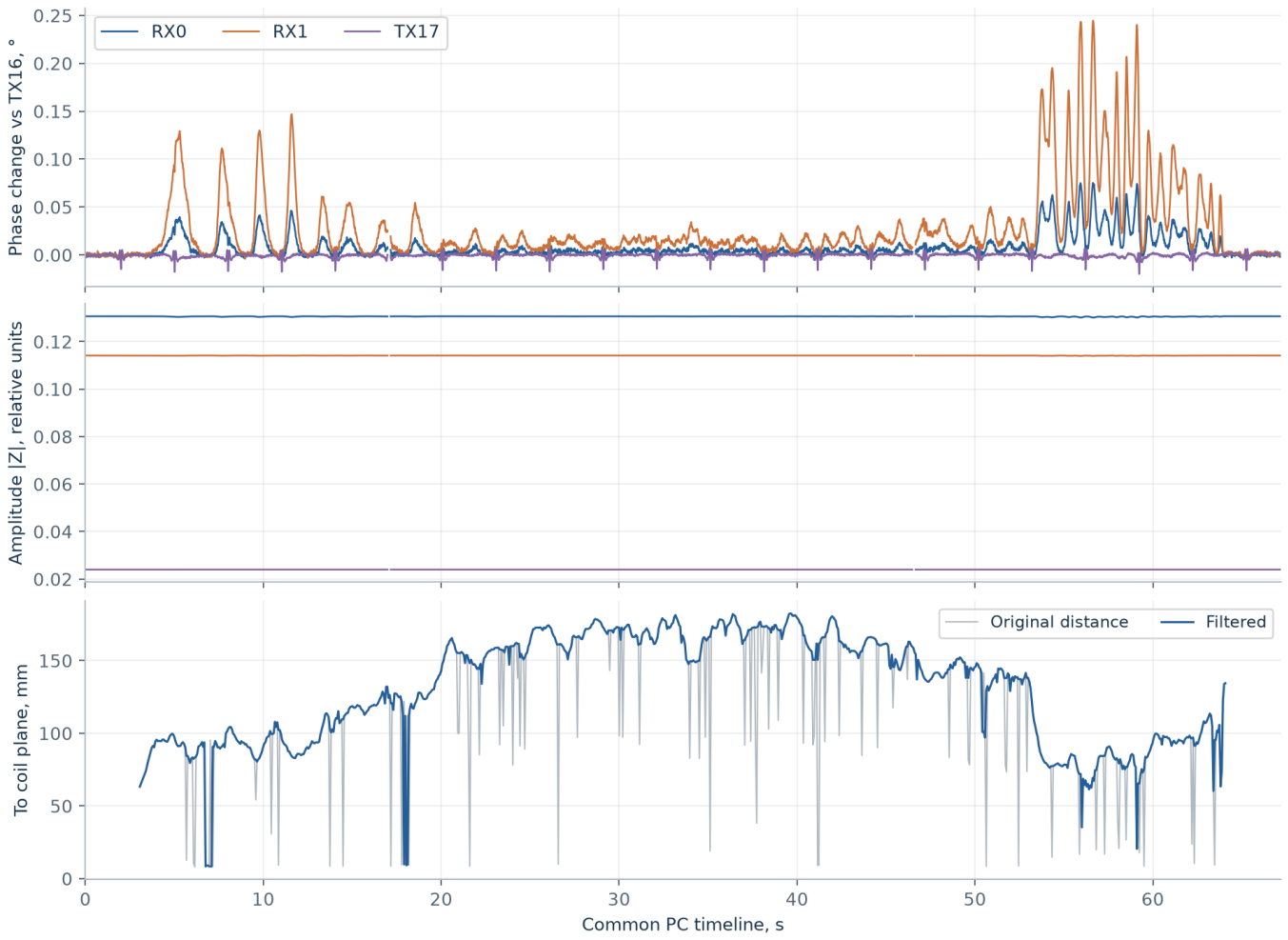
Each DSP input is mixed with 15.2 kHz sine/cosine references and passed through a fourth-order 100 Hz low-pass filter. Eight I/Q values are carried at 1500 / 3700 / 5900 / 8100 Hz: left = I, right = Q, pilot = 10300 Hz. PC decoding uses a 40 ms Hann window and 10 ms hop, yielding 100 sets/s. These are complex envelopes, not four full-bandwidth ADC streams.

Pulse check: all 20 expected plateaus were found. Across 396 settled frames, maximum absolute bias was 8.97×10^{-7} and I/Q RMS error ranged from 2.77×10^{-7} to 9.29×10^{-7}; ±100 ms around transitions was excluded. This validates transmission of prescribed values, not metal measurement accuracy.

60 s background: 5825/5997 valid sets, 123 rejected after the first second, and seven timing-discontinuity intervals. Gaps remain visible. A shared generator/ADC clock does not prevent capture losses on the PC.

11 NOVEMBER · GOLD RING 21 × 3 MM

Signal and target position

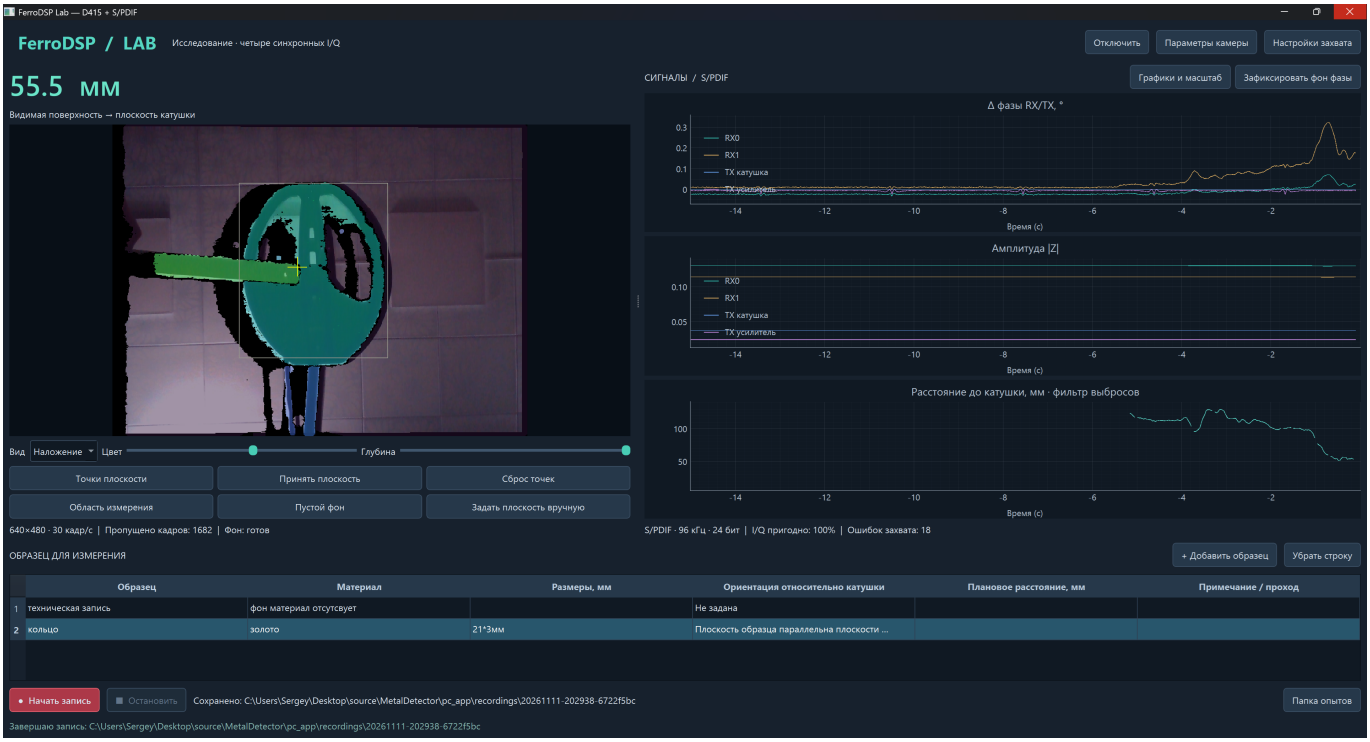


Ring plane parallel to the coil. D415: depth and color at 640 × 480, 15 fps. Recorded 6715 I/Q sets, 99.4937% valid, 999 camera frames and 905 complete observations. Maximum nearest signal/frame difference was 5.027 ms on the PC timeline. **Physical camera-to-audio latency was not calibrated.**

Distance refers to the nearest visible foreground surface and the coil plane; it may be a hand or holder rather than the ring center. A five-frame Hampel filter with threshold max(15 mm, 3 × 1.4826 MAD) corrected 78 outliers. Gaps were not filled; center-frame timestamps were preserved.

REPRODUCIBILITY AND NEXT CHECKS

Laboratory and findings



The author supplied this interface screenshot; its counters are cumulative and do not replace per-recording validation. Demonstration: H.264/AVC, 1280 × 640, 15 fps, 67.2 s. Open video.

Three predictions saved before reveal matched the identities of familiar objects: aluminium, copper and ring. Two trials differed from the instructed orientation. A magnetic caliper at approximately 250 mm received the saved answer “uncertain”. This is not an independent classifier accuracy evaluation.

Next algorithm: quality gating → complex normalization to TX16 → background estimation frozen during target events → ΔI/ΔQ features from both RX channels plus the TX16/TX17 ratio → evaluation by whole passes and unseen objects. Retain all four channels initially. Size, material and depth estimation require controlled geometry and a separate test set.

Numeric derivatives, source hashes, methodology and build script: report materials on GitHub. Original WAV and depth arrays remain in the local archive; published NPZ files reproduce these plots.